



ESOF 322

Software Engineering

LECTURE: 10

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# OUTLINE PR

- ⌘ Project Backlog
- ⌘ Semantic Versioning
- ⌘ Code Merge Strategy Guide

# PROJECT

**Backlog**

# MONTANA TECH MINER - PROJECT BACKLOG

## Current (DONE)

- ⚡ v0.1.0 Player movement + collision
- ⚡ v0.2.0 Tilemap + grid system
- ⚡ v0.2.5 Code Review + PR workflow training

# UPCOMING SPRINTS (NO BLOCKING DEPENDENCIES)

## v0.4.0 Stamina System Logic + Debug Output

- ⌘ Jump costs stamina
- ⌘ Food regenerates stamina
- ⌘ Movement/jump penalties when stamina is low
- ⌘ Debug text output

## v0.4.1 Stamina Bar Visualization

- ⌘ Blue stamina bar on HUD
- ⌘ Synchronized with v0.4.0 logic

# SEMANTIC

**Versioning**

# SEMANTIC VERSIONING GUIDE

## Overview

We use **Semantic Versioning** (SemVer) to track releases of the Montana Tech Miner project. This helps us communicate what changed and how significant the changes are.

v0.4.1

↓ ↓ ↓

| | └── PATCH: Bug fixes, small improvements

| └── MINOR: New features, backwards compatible

└── MAJOR: Breaking changes, major milestones

# OUR VERSIONING SCHEME

`v0.x.x` = Pre-Release / Alpha Phase

Since the game is under active development, all versions start with `0`. This means:

- ✂ Features may change or be removed.
- ✂ We're still experimenting with design.
- ✂ Breaking changes are acceptable in v0.x releases.

`v1.0.0` = First Stable Release

When we have a complete, playable game with all core features, we release `v1.0.0`.

# EXAMPLES FROM MONTANA TECH MINER

## Current Release History

- v0.1.0 → Player movement + basic collision
- v0.2.0 → Tilemap system + grid
- v0.2.5 → Code Review workflow training (minor polish)
- v0.3.0 → Render target scaling (if implemented)
- v0.4.0 → Stamina system logic + debug output
- v0.4.1 → Stamina bar visualization
- v0.5.0 → Food pickup mechanic
- v0.6.0 → Mining/digging system
- v0.6.5 → New tile types (STONE, COAL, CRYSTAL)
- v0.7.0 → Health system
- v0.8.0 → Win/lose conditions
- v0.9.0 → Polish (graphics, sounds, menu)
- v1.0.0 → First stable release

# Rules for Version Bumping 1/2

## PATCH (X.X.Z)

- ✖ Bug fixes in existing features
- ✖ Small optimizations
- ✖ Documentation updates
- ✖ No new functionality

### Example:

- ✖ v0.4.0 → v0.4.1: Fix stamina bar alignment bug

## MINOR (X.Y.X)

- ✖ New features added
- ✖ New tiles, new mechanics
- ✖ Backwards compatible with previous v0.x releases
- ✖ Existing code still works

### Example:

- ✖ v0.4.0 → v0.5.0: Add food pickup mechanic (new feature)
- ✖ v0.5.0 → v0.6.0: Add mining system (new feature)

## RULES FOR VERSION BUMPING 2/2

### MAJOR (X.x.x)

- ⚡ Significant overhaul
- ⚡ Breaking changes to existing systems
- ⚡ Game is now "complete enough" to move to v1.0.0

### Example:

- ⚡ v0.9.0 → v1.0.0: Game is complete and stable

# HOW TO CREATE A RELEASE TAG IN GIT

## 1. Ensure code is on main branch

```
git checkout main  
git pull origin main
```

## 2. Create an annotated tag

```
git tag -a v0.4.1 -m "Release: Stamina Bar Visualization"
```

## 3. Push tag to GitHub

```
git push origin v0.4.1
```

Or push all tags at once:

```
git push origin --tags
```

## 4. Verify tags on GitHub

Go to your repository → Releases tab. You should see the new tag listed.

# CODE

**Merge Strategy Guide**

# Overview

When multiple students implement the same feature (like Stamina System), we need a **merge strategy** to decide which code goes into `main` and how. This guide explains strategies used in modern software development.

## Context: Why We Need a Strategy

### The Problem:

- ✂ 4 students in Team Alpha all implement v0.4.0 (Stamina System).
- ✂ All 4 PRs work, but with slightly different approaches.
- ✂ All 4 PRs pass code review.
- ✂ We can't merge all 4 — that would create duplicate code.

### The Question:

How do we choose which code wins?

# Strategy 1: First One Wins (Simplest)

## HOW IT WORKS

### Timeline:

PR #10 (Mario) → Code review → Approved → MERGED to main

PR #11 (Luigi) → Code review → Approved → CLOSED (not merged)

PR #12 (Peach) → Code review → Approved → CLOSED (not merged)

PR #13 (Toad) → Code review → Approved → CLOSED (not merged)

## PROS

- ✂ Simple, fast
- ✂ No conflicts
- ✂ Clear winner

## CONS

- ✂ Unfair to other students (their work is discarded)
- ✂ You may lose good ideas from PR `#11`, `#12`, `#13`
- ✂ Demoralizing for students who wrote better code but were second

## WHEN TO USE

- ⌘ When time is critical
- ⌘ When one PR is objectively superior
- ⌘ In early development sprints

### Example Command

```
git checkout main  
git pull origin main  
git merge --squash alpha/feature/stamina_system/mario  
git commit -m "feat: Stamina system (v0.4.0) - mario"  
git push origin main  
git tag -a v0.4.0 -m "Stamina system complete"
```

## Strategy 2: Cherry-Picking

### HOW IT WORKS

Manually review all 4 PRs and **extract the best parts from each**, combining them into one optimal implementation.

PR #10 (mario):

- ✓ Good stamina drain logic
- ✓ Good variable naming
- ✗ Poor stamina regeneration

PR #11 (luigi):

- ✓ Excellent regeneration logic
- ✓ Clear comments
- ✗ Messy variable names

PR #12 (peach):

- ✓ Best debug output
- ✓ Good error handling
- ✓ Clean code structure

PR #13 (toad):

- ✗ Doesn't compile
- ✗ Incomplete

### Decision:

- ✗ Take drain logic from mario
  - ✗ Take regen logic from luigi
  - ✗ Take debug output from peach
  - ✗ Take error handling from peach
  - ✗ Create one clean, synthesized version
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# Implementation Steps

```
# 1. Create a new branch for synthesis
git checkout main
git pull origin main
git checkout -b feat/stamina_synthesis

# 2. Cherry-pick specific commits from mario
git cherry-pick mario_commit_hash_1
git cherry-pick mario_commit_hash_2

# 3. Cherry-pick specific commits from luigi
git cherry-pick luigi_commit_hash_1

# 4. Manually edit to combine best parts
# (Edit files to blend the approaches)

# 5. Create one clean synthesized commit
git add .
git commit -m "feat: Stamina system (v0.4.0) - Synthesized from team review"
```

## Combines:

- ✂ Drain logic from PR #10 (Mario)
- ✂ Regen logic from PR #11 (Luigi)
- ✂ Debug output from PR #12 (Peach)

Team consensus decision made during code review session."

# Merge to main

```
git checkout main  
git merge --squash feat/stamina_synthesis  
git push origin main  
git tag -a v0.4.0 -m "Stamina system complete"
```

## PROS

- ⌘ **Fair:** Acknowledges good work in all PRs
- ⌘ **Quality:** Final code is the best of all implementations
- ⌘ **Educational:** Students learn from each other's approaches
- ⌘ **Team morale:** Everyone contributed to final product

## CONS

- ⌘ Time-consuming (requires careful code review and synthesis)
- ⌘ Risk of introducing bugs during synthesis
- ⌘ Requires deep understanding of all 4 implementations

## WHEN TO USE

- ✚ When all PRs are good quality but slightly different
- ✚ In an educational setting (like this course)
- ✚ When combining different approaches makes sense

## EXAMPLE: COMPARISON TABLE

| Criterion       | mario | luigi | peach | toad | CHOOSE       |
|-----------------|-------|-------|-------|------|--------------|
| Drain logic     | ✓ ✓   | ✓     | ✓     | ✗    | mario        |
| Regen logic     | ✓     | ✓ ✓ ✓ | ✓     | ✗    | luigi        |
| Debug output    | ✓     | ✓     | ✓ ✓ ✓ | ✗    | peach        |
| Error handling  | ✓     | ✓     | ✓ ✓   | ✗    | peach        |
| Variable naming | ✓ ✓   | ✓     | ✓ ✓   | ✗    | mario        |
| Code comments   | ✓     | ✓ ✓   | ✓     | ✗    | luigi        |
| Compiles        | YES   | YES   | YES   | NO   | All but toad |

## Strategy 3: Squash and Merge (Git Technique)

### HOW IT WORKS

When merging a PR to `main`, combine all commits from the PR into **one single commit**.

Before (PR has 5 commits):

```
commit abc123: Add stamina variable  
commit def456: Add drain logic  
commit ghi789: Fix bug in regen  
commit jkl012: Add debug output  
commit mno345: Final polish
```

After squash merge to main:

```
commit xyz999: feat: Stamina system (v0.4.0)  
(contains all changes from abc123-mno345)
```

### WHY USE IT

- ✂ Keeps `main` branch history **clean and readable**
- ✂ Each PR = one logical commit
- ✂ Easier to revert a feature (just revert one commit)
- ✂ Easier to cherry-pick a feature (it's all in one commit)

# Git Command

```
git checkout main
git pull origin main

# Merge with --squash flag
git merge --squash feature/alpha/stamina_system/mario

# Create one commit
git commit -m "feat: Stamina system (v0.4.0) - mario"
```

## Implements:

- ❌ Stamina drain on jump
- ❌ Stamina regeneration from food
- ❌ Movement/jump penalties when low"

```
git push origin main
```

## PROS

- ❌ Clean history on `main`
- ❌ Easy to track logical changes
- ❌ PR still visible in GitHub (even after merge)

## CONS

- ❌ Loses individual commit history
- ❌ Can't see step-by-step development

## WHEN TO USE

- ❌ Always (recommended for most projects)
- ❌ Especially when `main` branch is critical

## Strategy 4: Rebase and Merge

### HOW IT WORKS

Instead of creating a merge commit, **replay all PR commits on top of main.**

Before:

```
main: A ← B ← C  
PR: D ← E ← F
```

After (rebase):

```
main: A ← B ← C ← D ← E ← F
```

### PROS

- ✂ Linear, clean history (no merge commits)
- ✂ Easier to bisect (find which commit broke something)
- ✂ Each commit is testable independently

### CONS

- ✂ Rewrites commit history (destructive)
- ✂ Can cause conflicts if not careful
- ✂ Less clear that changes were reviewed as a unit

## Git Command

```
git checkout feature/alpha/stamina_system/mario  
git rebase origin/main  
  
git checkout main  
git merge --ff-only feature/alpha/stamina_system/mario  
git push origin main
```

### WHEN TO USE

- ✂ In projects with linear development workflows
- ✂ When individual commits are well-tested
- ✂ Less common in educational settings

THANK

*You*